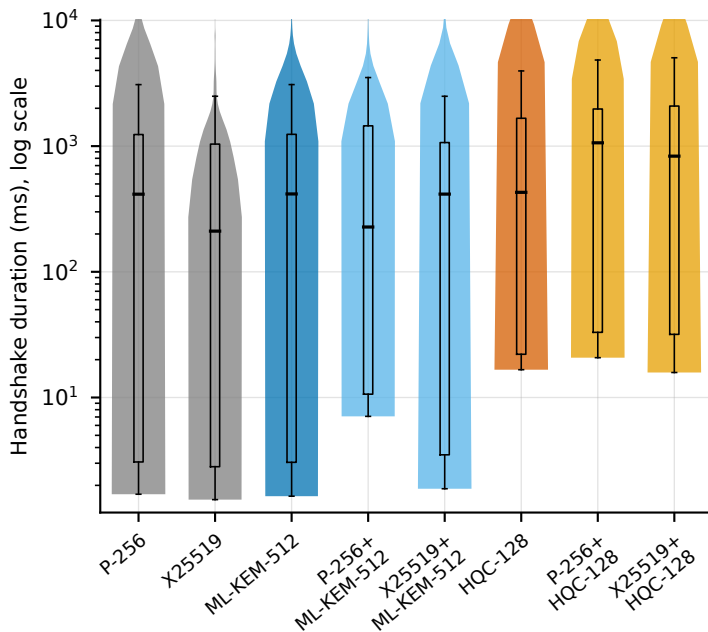


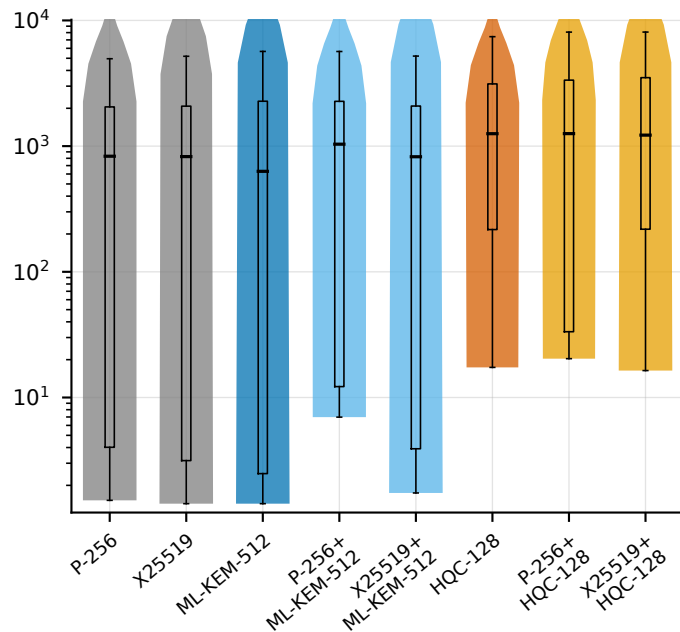
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	2.35k	629.3	214.06k	3.24k	476.9	224.09k
X25519	741.7	220.1	26.80k	4.49k	569.0	370.89k
ML-KEM-512	1.73k	442.5	107.74k	4.53k	592.2	456.63k
P-256+ML-KEM-512	1.43k	400.6	107.38k	3.70k	421.2	215.94k
X25519+ML-KEM-512	1.66k	617.3	216.59k	4.16k	668.4	461.35k
HQC-128	3.59k	622.9	458.98k	4.02k	339.6	217.20k
P-256+HQC-128	3.65k	580.9	336.89k	5.26k	360.6	228.33k
X25519+HQC-128	3.36k	642.0	461.04k	6.21k	434.0	457.06k